Dark Matter Profile Differentiation in SPARC Galaxies: Inner Velocity Slopes, Core Radii, and Baryonic Correlations

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1 Introduction

One of the most striking tensions in modern astrophysics emerges from galactic rotation curves: the observed orbital velocities of stars and gas at large galactic radii remain roughly constant rather than declining, as Keplerian mechanics predicts from the visible mass distribution [1]. This mass discrepancy implies the existence of extended, roughly spherical dark matter halos surrounding galaxies, contributing to the dominant gravitational potential at large radii [2].

Cold dark matter (CDM) N-body simulations make a specific prediction about the structure of these halos: density should rise steeply toward galactic centers, following a cuspy $\rho \propto r^{-1}$ profile described by the Navarro-Frenk-White (NFW) model [3]. However, observed rotation curves of late-type and dwarf galaxies frequently favor shallower, cored central density distributions, better described by the pseudo-isothermal (ISO) profile, in which the central density flattens to a finite constant [5]. This discrepancy between simulation and observation — the core-cusp problem — represents one of the most persistent open questions in small-scale structure formation. One of the most persistent tensions in modern cosmology is the core-cusp problem: N-body simulations of cold dark matter (CDM) consistently predict that dark matter density profiles should rise steeply toward galactic centers, following a cuspy $\rho \propto r^{-1}$ behavior described by the Navarro-Frenk-White (NFW) profile [3]. However, observed rotation curves of late-type and dwarf galaxies frequently favor shallower, cored density distributions, better described by the pseudo-isothermal (ISO) profile, in which the central density flattens to a finite constant [5].

This discrepancy has motivated two broad classes of explanations. The first involves baryonic feedback — supernova explosions and stellar winds repeatedly drive gas outflows that dynamically heat and redistribute dark matter, transforming an initial cusp into a core over time [7]. The second questions the cold dark matter paradigm itself, proposing alternatives such as warm dark matter or self-interacting dark matter [4]. Distinguishing between these explanations requires identifying which galaxy properties correlate with core or cusp behavior; in other words NFW or ISO profiles across large, well-characterized samples.

The Spitzer Photometry and Accurate Rotation Curves (SPARC) database provides an ideal dataset for this investigation, offering high-quality rotation curve decompositions for 175 late-type galaxies [9]. In this work, we fit both NFW and ISO dark matter profiles to the dark matter velocity component of each galaxy, apply Mann-Whitney U statistical tests to identify which observable properties best distinguish core from cusp galaxies, and examine how baryonic fraction relates to fitted core radii. We find that the inner slope of the dark matter velocity curve and the isothermal core radius are the strongest differentiators, and that baryonic fraction correlates strongly with core size among ISO-preferred galaxies, consistent with the baryonic feedback hypothesis.

2 Data and Methods

2.1 SPARC Dataset

This study uses rotation curve data from the SPARC database [9], which provides high-resolution kinematic measurements for 175 late-type galaxies. Each galaxy file contains the following columns:

Column	**Units**	**Description**
R	kpc	Galactocentric radius
$V_{\rm obs}$	km/s	Observed circular velocity
σ_V	km/s	Uncertainty on $V_{\rm obs}$
$V_{\rm gas}$	km/s	Velocity contribution from gas (HI + He)
$V_{\rm disk}$	km/s	Velocity contribution from stellar disk
$V_{\rm bul}$	km/s	Velocity contribution from bulge
$\Sigma_{\rm disk}$	$L_\odot/\mathrm{pc}^2$	Stellar disk surface brightness
$\Sigma_{\rm bul}$	$L_\odot/\mathrm{pc}^2$	Bulge surface brightness

Table 1: SPARC data columns available per galaxy. Columns used in this analysis are R, $V_{\rm obs}$, σ_V, $V_{\rm gas}$, $V_{\rm disk}$, and $V_{\rm bul}$. Surface brightness columns $\Sigma_{\rm disk}$ and $\Sigma_{\rm bul}$ are provided but not used in the primary fitting pipeline.

Of these, R, $V_{\rm obs}$, σ_V, $V_{\rm gas}$, $V_{\rm disk}$, and $V_{\rm bul}$ are used directly in this analysis. The total baryonic velocity is computed as:

$$V_{\rm bar} = \sqrt{V_{\rm gas}^2 + V_{\rm disk}^2 + V_{\rm bul}^2} \tag{1}$$

and the dark matter velocity component is extracted via quadrature subtraction:

$$V_{\rm DM} = \sqrt{V_{\rm obs}^2 - V_{\rm bar}^2} \tag{2}$$

Galaxies where $V_{\rm obs}^2 - V_{\rm bar}^2 \leq 0$ at most radii — indicating baryon-dominated kinematics with insufficient dark matter signal — were excluded from profile fitting. In total, 25 galaxies were excluded in the analysis pipeline, leaving 150 galaxies in the final sample.

2.2 Dark Matter Density Profiles

Two spherically symmetric dark matter density profiles were fitted to each galaxy. The NFW profile [3] predicts a cuspy central density:

$$\rho_{\rm NFW}(r) = \frac{\rho_0}{(r/r_s)(1 + r/r_s)^2} \tag{3}$$

where ρ_0 is a characteristic density and r_s is the scale radius. This profile diverges as $r \to 0$, producing a central density cusp.

The pseudo-isothermal (ISO) profile instead predicts a cored central density:

$$\rho_{\rm ISO}(r) = \frac{\rho_0}{1 + (r/r_c)^2} \tag{4}$$

where r_c is the core radius. As $r \to 0$, $\rho_{\rm ISO} \to \rho_0$, flattening to a finite constant rather than diverging. The enclosed mass $M(r)$ is obtained by integrating the density profile over a sphere of radius r:

$$M(r) = 4\pi \int_0^r \rho(r')\, r'^2\, dr' \tag{5}$$

For the NFW profile this integral has an the solution:

$$M_{\rm NFW}(r) = 4\pi\rho_0 r_s^3 \left[\ln\left(1 + \frac{r}{r_s}\right) - \frac{r/r_s}{1 + r/r_s}\right] \tag{6}$$

For the ISO profile, we evaluate $M_{\rm ISO}(r)$ numerically using cumulative trapezoidal integration on a shared radial grid, then interpolate to the observed radii. The corresponding circular velocity for each profile is:

$$V(r) = \sqrt{\frac{G \cdot M(r)}{r}} \tag{7}$$

where $M(r)$ is the cumulative mass enclosed within radius r, computed analytically for NFW and numerically for ISO.

2.3 Rotation Curves and Density Profiles

For each galaxy, the observed rotation curve and fitted model velocities are plotted as a function of galactocentric radius. As a representative example, Figure 1 and Figure 2 show results for galaxy D631-7, a late-type dwarf galaxy from the SPARC sample for which the ISO profile provides a significantly better fit than NFW ($\chi^2_{\rm NFW} = 3.66$, $\chi^2_{\rm ISO} = 0.86$). Figure 1 shows a representative example for galaxy D631-7, displaying $V_{\rm obs}$, $V_{\rm bar}$, and the extracted $V_{\rm DM}$ alongside the best-fit NFW and ISO model curves. The corresponding dark matter density profiles $\rho(r)$ derived from each fit are shown on a log-log scale in Figure 2, where the core-cusp distinction is most visually apparent: the ISO profile flattens at small r while the NFW profile continues to rise.

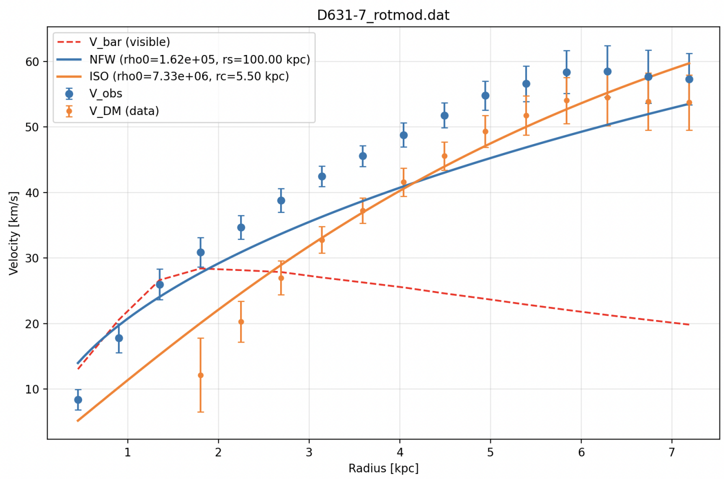

Figure 1: Rotation curve decomposition for D631-7. Blue points show $V_{\rm obs}$ with error bars, the red dashed line shows $V_{\rm bar}$, orange points show the extracted $V_{\rm DM}$, and solid lines show the best-fit NFW (blue) and ISO (orange) model curves.

2.4 Profile Fitting and Model Selection

Both profiles were fitted to $V_{\rm DM}(r)$ using nonlinear least-squares minimization via `scipy.optimize.curve_fit`, with per-point weighting by the propagated dark matter velocity uncertainty $\sigma_{V_{\rm DM}}$:

$$\sigma_{V_{\rm DM}} = \frac{V_{\rm obs}}{V_{\rm DM}} \cdot \sigma_V \tag{8}$$

Model performance was evaluated using the weighted reduced chi-squared:

$$\chi^2_{\rm red} = \frac{1}{N-2} \sum_{i=1}^{N} \left(\frac{V_{{\rm DM},i} - V_{{\rm model},i}}{\sigma_{V_{{\rm DM},i}}} \right)^2 \tag{9}$$

where N is the number of valid data points. A galaxy was classified as ISO-preferred if $\ln(\chi^2_{\rm NFW}/\chi^2_{\rm ISO}) > 0.33$, NFW-preferred if $\ln(\chi^2_{\rm NFW}/\chi^2_{\rm ISO}) < -0.33$, and ambiguous (Neither) otherwise. The threshold of 0.33 was chosen empirically, corresponding to one profile fitting approximately 39% better than the other ($e^{0.33} \approx 1.39$), defining a grey zone where neither profile fits decisively better. Galaxies with fewer than 4 valid $V_{\rm DM}$ data points after exclusion of non-physical values were also excluded, as an insufficient number of points cannot reliably constrain a two-parameter model. In total, 25 galaxies were excluded on these grounds, leaving 150 galaxies in the final sample.

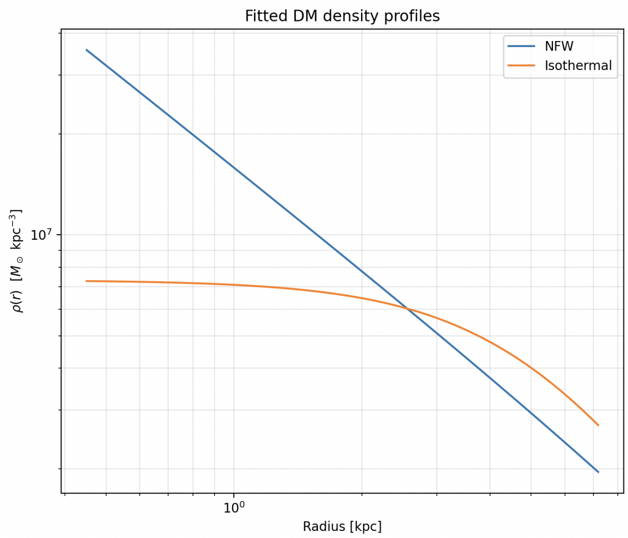

Figure 2: Dark matter density profiles on a log-log scale for D631-7. The NFW profile (blue) rises steeply toward the center, while the ISO profile (orange) flattens to a finite core density. This distinction defines the core-cusp problem.

2.5 Mann-Whitney U Statistical Analysis

To identify which galaxy properties most strongly differentiate ISO-preferred from NFW-preferred galaxies, we applied the Mann-Whitney U test independently to each of 11 variables. The test statistic is defined as:

$$U = \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} S(x_i, y_j), \quad S(x_i, y_j) = \begin{cases} 1 & \text{if } x_i > y_j \\ 0.5 & \text{if } x_i = y_j \\ 0 & \text{if } x_i < y_j \end{cases} \tag{10}$$

where n_1 and n_2 are the sizes of the ISO and NFW groups respectively, and x_i, y_j are values of the variable being tested. Unlike a t-test, the Mann-Whitney U test makes no assumption of normality, making it appropriate for the skewed distributions typical of astrophysical parameters. A two-sided p-value was computed for each variable, and a result was considered statistically significant at $p < 0.05$.

Of the 11 variables tested, only two were statistically significant: the inner slope of V_{DM} at small radii ($p = 0.003$) and the ISO core radius r_c ($p = 0.028$). All other variables including peak observed velocity, maximum radius, baryonic fraction, and central surface brightness showed no statistically significant difference between the two groups, with a much greater p value, indicating that inner slope and core radius are the key physical differentiators between core and cusp galaxies in this sample.

The inner slope was computed directly from the raw V_{DM} data by performing a linear regression of $\ln V_{\mathrm{DM}}$ against $\ln r$ over the innermost three valid data points, giving the slope estimate:

$$\alpha = \frac{\sum_i (\ln r_i - \overline{\ln r})(\ln V_i - \overline{\ln V})}{\sum_i (\ln r_i - \overline{\ln r})^2} \tag{11}$$

This serves as a numerical approximation to the logarithmic derivative $\frac{d \ln V_{\mathrm{DM}}}{d \ln r}\Big|_{r \to 0}$, which characterizes the inner slope of the dark matter velocity profile in the limit of small radii. For a purely cored profile, $V_{\mathrm{DM}} \propto r$ near the center, giving $\alpha \to 1$. For a cuspy NFW profile, the central mass distribution produces a steeper rise with $\alpha > 1$. The Mann-Whitney result ($p = 0.003$) confirms that this local logarithmic slope is the single strongest model-independent differentiator between ISO and NFW galaxies in the SPARC sample.

2.6 Baryonic Matter Analysis

To investigate whether baryonic content influences dark matter profile shape, we define the mean baryonic fraction for each galaxy through the radial average as:

$$\bar{f}_{\mathrm{bar}} = \left\langle \frac{V_{\mathrm{bar}}^2(r)}{V_{\mathrm{obs}}^2(r)} \right\rangle_r \tag{12}$$

where the average is taken over all valid radial data points. This quantity represents the fraction of the total gravitational potential at each radius that is accounted for by baryonic matter, averaged across the galaxy. A value of $\bar{f}_{\mathrm{bar}} \approx 1$ indicates a baryon-dominated galaxy with little dark matter contribution, while $\bar{f}_{\mathrm{bar}} \approx 0$ indicates a dark matter-dominated system.

Galaxies were split into high and low baryonic fraction groups using the median $\bar{f}_{\mathrm{bar}}$ of each profile group as the threshold (ISO median: 0.527, NFW median: 0.411). For each group, we then counted how many galaxies fell into each combination of baryonic fraction and profile indicator — either inner slope α or core radius r_c — using the respective group medians as thresholds.

Among ISO-preferred galaxies, the distribution across baryonic fraction and core radius reveals a striking diagonal pattern. Of the 91 ISO galaxies with valid data, 43% show both high $\bar{f}_{\mathrm{bar}}$ and large r_c (above the median of 3.78 kpc), while 44% show both low $\bar{f}_{\mathrm{bar}}$ and small r_c. The opposite combinations — high baryonic fraction with small core, and low baryonic fraction with large core — each account for only 7% of the sample. This near-symmetric diagonal split is expressed as:

$$P(\text{large } r_c \mid \text{high } \bar{f}_{\mathrm{bar}}) \gg P(\text{large } r_c \mid \text{low } \bar{f}_{\mathrm{bar}}) \tag{13}$$

indicating a strong positive association between baryonic content and core size. Galaxies with more baryonic matter tend to have larger dark matter cores, while galaxies with less baryonic matter tend to have smaller cores. This pattern is consistent with the baryonic feedback hypothesis [7]: repeated supernova-driven gas outflows in baryon-rich systems transfer energy to dark matter particles, gradually flattening a central density cusp into an extended core. The rarity of the opposite combinations (7% each) further suggests this is not a random distribution but a physically motivated correlation.

No analogous pattern was observed for inner slope α among either ISO or NFW galaxies, where all four baryonic-slope combinations appeared with roughly equal frequency. This suggests that while $\bar{f}_{\mathrm{bar}}$ predicts the *size* of a dark matter core, it does not predict the *shape* of the central velocity rise independently of the profile classification.

3 Results

3.1 Reduced Chi-Squared Profile Classification

Applying the weighted reduced chi-squared fitting procedure and logarithmic ratio classification threshold across all 175 SPARC galaxies, we obtain the following profile classifications:

Classification	**Count**	**Percentage**
ISO-preferred (core)	91	52.0%
NFW-preferred (cusp)	15	8.6%
Neither (ambiguous)	44	25.1%
Failed (insufficient data)	25	14.3%
Total	175	100%

Table 2: Profile classification results across 175 SPARC galaxies. Failed galaxies were excluded due to insufficient valid V_{DM} data points, typically arising in baryon-dominated systems where $V_{\mathrm{obs}}^2 - V_{\mathrm{bar}}^2 \leq 0$ at most radii.

The ISO profile is preferred in the majority of galaxies where a decisive classification was possible, with NFW preferred in only 15 cases. The large Neither category reflects galaxies where both profiles fit comparably well, suggesting the data do not strongly constrain the central density shape in these cases. The dominance of ISO over NFW classifications is consistent with prior observational studies finding cored profiles in late-type and dwarf galaxies [5].

3.2 Mann-Whitney U Test Results

Applying the Mann-Whitney U test across 11 galaxy properties, only two variables showed statistically significant differences between ISO-preferred and NFW-preferred galaxies at the $p < 0.05$ level: the inner dark matter velocity slope α and the isothermal core radius r_c.

The inner slope α is significantly higher in ISO-preferred galaxies (median $\alpha_{\mathrm{ISO}} = 1.078$) compared to NFW-preferred galaxies (median $\alpha_{\mathrm{NFW}} = 0.546$), with a correlation strength of 99.7% ($p = 0.003$). This result is physically meaningful: ISO galaxies exhibit a more gradual rise in V_{DM} toward the galactic

Variable	Correlation	p-value	Significant	Higher in
α (inner slope)	99.7%	0.003	Yes	ISO
r_c (core radius)	97.2%	0.028	Yes	ISO
$\bar{f}_{\mathrm{bar}}$	92.0%	0.080	No	ISO
χ^2_{ISO}	87.0%	0.130	No	NFW
$V_{\mathrm{obs,peak}}$	20.0%	0.800	No	ISO
Σ_{disk}	16.5%	0.835	No	ISO
r_{max}	11.2%	0.888	No	ISO

Table 3: Mann-Whitney U test results for selected variables, ranked by correlation strength. Only α and r_c are statistically significant. Remaining variables showed no significant separation between ISO and NFW groups.

center, consistent with a flat central density core, while NFW galaxies show a steeper inner velocity rise consistent with a diverging central density cusp. Crucially, α is computed directly from the observed V_{DM} data without reference to any model, making it a model-independent observational indicator of the core-cusp distinction.

The isothermal core radius r_c is also significantly higher in ISO galaxies (median $r_{c,\mathrm{ISO}} = 3.78$ kpc) than in NFW galaxies (median $r_{c,\mathrm{NFW}} = 1.69$ kpc, $p = 0.028$). This indicates that when the ISO profile wins, it does so with a genuinely extended core rather than a marginally flattened one — the fitted core radii are physically large and not merely a fitting artifact.

Together, these two results present a consistent picture that directly challenges the predictions of cold dark matter (CDM) N-body simulations. Standard CDM predicts that dark matter halos should universally follow NFW-like cuspy profiles [3], yet the majority of galaxies in this sample not only prefer the ISO profile statistically, but also show lower inner velocity slopes and larger core radii — both independently pointing toward genuinely cored dark matter distributions. The fact that variables such as peak velocity, maximum radius, and central surface brightness show no significant separation between the two groups further suggests that the core-cusp distinction is not driven by galaxy size or luminosity, but reflects a more fundamental difference in dark matter halo structure.

3.3 Baryonic Fraction and Core Radius Correlation

Among the 91 galaxies preferred by ISO, we find a pronounced diagonal pattern between the mean baryonic fraction $\bar{f}_{\mathrm{bar}}$ and the fitted core radius r_c, shown in Figure 3.

The distribution is strikingly non-uniform. The two dominant categories are high $\bar{f}_{\mathrm{bar}}$ with large r_c (43%) and low $\bar{f}_{\mathrm{bar}}$ with small r_c (44%), while the cross combinations — high baryonic fraction with small core and low baryonic fraction with large core — each represent only 7% of the sample. This strong diagonal split indicates that baryonic fraction and core radius are not independent among

ISO galaxies: galaxies with more baryonic matter systematically tend to host larger dark matter cores, and galaxies with less baryonic matter tend to host smaller ones.

This pattern has direct implications for the core-cusp problem. The observed correlation between $\bar{f}_{\text{bar}}$ and r_c provides observational support for the baryonic feedback mechanism [7], in which repeated cycles of star formation and supernova-driven gas outflows transfer energy to dark matter particles in the central halo. In baryon-rich galaxies, this process is more vigorous and sustained, gradually expanding the dark matter core to larger radii. In baryon-poor, dark matter-dominated galaxies, this mechanism is less effective, and the core remains compact or absent. The rarity of the cross combinations (7% each) is particularly notable — it suggests that large cores rarely appear without substantial baryonic content, and that high baryonic fractions are commonly associated with extended cores. This pattern is consistent with, but does not by itself prove a causal connection.

No analogous pattern was observed among NFW-preferred galaxies, where baryonic fraction and inner slope were distributed approximately equally across all four combination categories. This asymmetry between ISO and NFW galaxies further reinforces the interpretation that the $\bar{f}_{\text{bar}}$–r_c correlation is physically meaningful and specific to galaxies that have undergone core formation, rather than a general property of the sample.

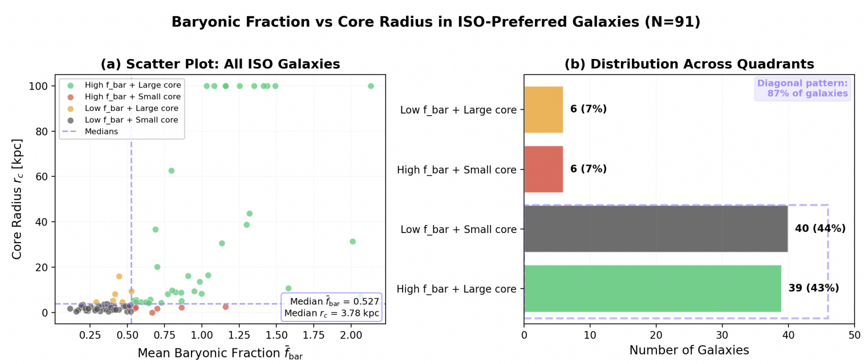

Figure 3: Correlation between baryonic fraction and core radius among ISO-preferred galaxies. **(a)** Scatter plot of all 91 ISO galaxies showing clear clustering in opposite diagonal quadrants. Dashed lines indicate group medians ($\bar{f}_{\text{bar}} = 0.527$, $r_c = 3.78$ kpc). **(b)** Distribution across quadrants defined by these medians, revealing that 87% of galaxies fall into the two diagonal categories (high baryonic fraction with large cores, or low baryonic fraction with small cores), consistent with the baryonic feedback hypothesis.

4 Conclusion

This study applied NFW and pseudo-isothermal dark matter profile fitting to 175 late-type galaxies from the SPARC database, using weighted reduced chi-squared minimization and Mann-Whitney U statistical analysis to identify the physical properties that most strongly distinguish cored from cuspy dark matter distributions.

The results alone present a challenge to standard cold dark matter predictions: ISO is preferred in 52% of galaxies where a decisive classification was possible, while NFW is preferred in only 8.6%. If CDM N-body simulations were an accurate description of galaxy-scale dark matter structure, NFW profiles should dominate. Instead, the observed rotation curves of late-type galaxies in the SPARC sample overwhelmingly favor cored distributions, reinforcing a tension that has persisted in the literature for decades [5].

The Mann-Whitney analysis sharpens this picture considerably. Of 11 galaxy properties tested, only the inner dark matter velocity slope α ($p = 0.003$) and the isothermal core radius r_c ($p = 0.028$) show statistically significant differences between ISO and NFW galaxies. Critically, α is a model-independent quantity computed directly from the observed $V_{\rm DM}$ data — it does not assume either profile. Its strong separation between the two groups means that the core-cusp distinction is not merely a consequence of which model is fitted, but is encoded in the data itself. ISO galaxies have shallower central velocity rises; NFW galaxies have steeper ones. This is the observational signature of the core-cusp problem expressed directly in velocity space, without model dependence.

The baryonic fraction analysis adds a physical dimension to these findings. Among ISO-preferred galaxies, we find a strong diagonal correlation between $\bar{f}_{\rm bar}$ and r_c: 43% of ISO galaxies show both high baryonic fraction and large core radius, while 44% show both low baryonic fraction and small core radius. The opposite combinations each represent only 7% of the sample. This pattern is absent in NFW galaxies, making it specific to systems that have undergone core formation. The physical interpretation is baryonic feedback [7]: in baryon-rich galaxies, repeated supernova-driven outflows transfer enough energy to the dark matter halo to flatten a cusp into an extended core. In baryon-poor galaxies, this mechanism is insufficient, and the core either never forms or remains compact. The data are consistent with this picture in this sample — the galaxies with the most baryonic matter have the largest cores, and those with the least have the smallest.

Ultimately, the two main findings of this work — that α and r_c are the strongest differentiators between ISO and NFW galaxies, and that $\bar{f}_{\rm bar}$ predicts core size among ISO galaxies — point toward a coherent physical narrative. The core-cusp problem appears unlikely to be only a fitting ambiguity in this dataset; it is a structural difference between galaxies, visible in model-independent velocity data, and correlated with the amount of baryonic matter available to drive feedback. Whether baryonic feedback alone is sufficient to fully resolve the tension with CDM predictions, or whether modifications to the dark matter paradigm are ultimately required, remains an open question. But the results presented here suggest that the answer lies in the complex interplay between visible and dark matter, and that the inner slope of the dark matter velocity curve may be the clearest observational window into that relationship.

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